Sliced Ollivier-Ricci Curvature: A Provable Spectral Lower Bound for Discrete Optimal Transport

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Abstract

Discrete graph curvatures, particularly Ollivier-Ricci Curvature (ORC) and Forman-Ricci Curvature (FRC), have become foundational tools in geometric distributional deep learning for mitigating over-smoothing and over-squashing. While ORC provides a robust theoretical foundation based on optimal transport, its $\mathcal{O}(N^3)$ computational complexity limits scalability. Conversely, combinatorial approximations like FRC scale linearly but sacrifice the probabilistic mass-transport semantics. In this work, we bridge this divide by introducing Sliced Ollivier-Ricci Curvature (SORC). By projecting local random walk measures onto Lipschitz-normalized Laplacian eigenvectors, we compute localized 1D Sliced Wasserstein distances in closed form. We prove that SORC mathematically lower-bounds exact ORC via an optimal transport coupling formulation. We validate this on synthetic stochastic block models, where SORC preserves high topological correlation and achieves a dramatic speedup over exact ORC. However, in evaluations spanning large real-world recommendation graphs, SORC achieves only moderate correlation in downstream link prediction tasks, failing to fully match purely combinatorial approximations. Code is provided in the supplementary material for anonymous review.

1 Introduction

The geometry of message passing in Graph Neural Networks (GNNs) is fundamentally dictated by the underlying graph topology. Recent advances in geometric deep learning have established that negative curvature induces over-squashing due to bottleneck structures, while positive curvature exacerbates over-smoothing. Ollivier-Ricci Curvature (ORC) mitigates these issues by quantifying neighborhood overlap via the 1-Wasserstein (W_1) distance, but requires solving an expensive linear program for every edge.

To overcome this computational barrier, combinatorial surrogates such as Augmented Forman-Ricci Curvature (AFRC) count local cycles to approximate the optimal transport structure [2]. Other spectral approaches, such as Lin-Lu-Yau curvature [4], approximate the transport plan but remain computationally demanding. Meanwhile, Sliced Wasserstein distances have seen broad adoption in continuous machine learning domains [5], leveraging the fact that 1D Wasserstein distances can be computed by simply sorting the empirical measures. However, sliced optimal transport remains largely under-explored for discrete graphs due to the lack of canonical projection axes.

In this paper, we propose a spectral optimal transport approach: Sliced Ollivier-Ricci Curvature (SORC). We demonstrate that by carefully normalizing the eigenvectors of the graph Laplacian to be 1-Lipschitz, the 1D Wasserstein distance along these spectral projections yields a mathematically provable lower bound on exact ORC, computable in a fraction of the time.

2 Related Work

Graph Curvature and Optimal Transport. The discrete analog of Ricci curvature was introduced by Ollivier [1] based on the optimal transport cost between random walks. This geometric lens has proven vital in graph neural networks for identifying and rewiring structural bottlenecks that cause over-squashing. However, computing the Earth Mover's Distance between all adjacent neighborhoods requires solving $|E|$ linear programs, severely limiting scalability on large graphs.

Combinatorial Approximations. To bypass optimal transport solvers, researchers have turned to combinatorial alternatives. Forman-Ricci Curvature (FRC) calculates curvature based on node degrees and simple topological motifs. Balanced Forman Curvature, popularized by Topping et al. [6] as a diagnostic tool for over-squashing bottlenecks, demonstrates the power of combinatorial rewiring. Extensions like Augmented Forman-Ricci Curvature (AFRC) [2] account for higher-order structures like 4-cycles to better approximate ORC. While these methods achieve $\mathcal{O}(|E|)$ runtime, they heuristically count motifs rather than explicitly solving a mass-preservation problem, resulting in weaker correlations with the true transport geometry. Lin-Lu-Yau curvature [4] modifies the random walk definition to avoid lazy random walks, but still relies on computational matchings. We position SORC as a spectral alternative to these combinatorial approximations.

Sliced Wasserstein Distances. Sliced Optimal Transport [5] mitigates the curse of dimensionality by projecting high-dimensional measures onto 1D lines, where the Wasserstein distance admits a closed-form solution via the inverse CDF. While successful in continuous image processing and generative modeling, its application to discrete graph topologies requires identifying a set of projection axes that respect the graph's shortest-path metric. Our work introduces Lipschitz-normalized Laplacian eigenvectors as the natural axes for sliced transport on graphs.

3 Definitions and Theoretical Results

Let $G = (V, E)$ be an unweighted, undirected graph. For any vertex u, we define the local random walk measure with idleness $\alpha = 0.5$ as $m_u^\alpha = \alpha\delta_u + (1-\alpha)\text{Unif}(N(u))$. The standard ORC along edge $\{u, v\}$ is $\kappa_{\text{ORC}}(u, v) = 1 - W_1(m_u^\alpha, m_v^\alpha)/d(u, v)$.

3.1 Lipschitz Normalization of Eigenvectors

Let ϕ_i (for $i \geq 2$) be a non-trivial eigenvector of the graph Laplacian $L = D - A$, explicitly excluding the trivial constant vector ϕ_1 corresponding to $\lambda_1 = 0$ to prevent division by zero. To utilize ϕ_i as a test function, it must be 1-Lipschitz with respect to the graph shortest-path metric $d(x, y)$. We compute the maximum edge gradient $\Delta_{\max}(\phi_i) = \max_{\{x,y\}\in E} |\phi_i(x) - \phi_i(y)|$, and define the normalized eigenvector $f_i = \phi_i/\Delta_{\max}(\phi_i)$.

Theorem 1. *Let $SW_k(m_u, m_v) = \max_{1\leq i\leq k} W_1(f_{i\#}m_u, f_{i\#}m_v)$ be the maximum 1D Sliced Wasserstein distance along the first k Lipschitz-normalized Laplacian eigenvectors. Then, $SW_k(m_u, m_v) \leq W_1(m_u, m_v)$.*

Proof. It is a fundamental property of optimal transport that pushforward measures under a 1-Lipschitz map are non-expansive with respect to the 1-Wasserstein distance [3]. Let π be any optimal coupling of (m_u, m_v) on the graph such that $\int d(x, y)d\pi(x, y) = W_1(m_u, m_v)$.

By construction, each normalized eigenvector $f_i = \phi_i/\Delta_{\max}(\phi_i)$ is 1-Lipschitz with respect to the graph metric d. Therefore, for any $x, y \in V$, we have $|f_i(x) - f_i(y)| \leq d(x, y)$. The pushforward $(f_i \times f_i)_\#\pi$ represents a valid transport plan between the 1D projected measures $f_{i\#}m_u$ and $f_{i\#}m_v$ on $\mathbb{R}$.

The cost of this 1D coupling is strictly bounded by the cost of the graph coupling:

$$
\begin{aligned}
W_1(f_{i\#}m_u, f_{i\#}m_v) &\leq \int_{\mathbb{R}\times\mathbb{R}} |x' - y'| d((f_i \times f_i)_{\#}\pi)(x', y') \\
&= \int_{V\times V} |f_i(x) - f_i(y)| d\pi(x, y) \\
&\leq \int_{V\times V} d(x, y) d\pi(x, y) = W_1(m_u, m_v)
\end{aligned}
$$

Since this bound holds independently for every projection axis f_i, taking the maximum over the k directions preserves the inequality, yielding $\mathrm{SW}_k(m_u, m_v) \leq W_1(m_u, m_v)$. □

Corollary 1. *The Sliced Ollivier-Ricci Curvature $\kappa^{(k)}_{SORC}(u, v) = 1 - SW_k(m^\alpha_u, m^\alpha_v)/d(u, v)$ guarantees that $\kappa^{(k)}_{SORC}(u, v) \geq \kappa_{ORC}(u, v)$.*

While the bound monotonically tightens as $k \to n$, the maximum over the spectral axes limits the exact recovery of W_1 to cases where the optimal coupling perfectly aligns with a single eigenvector. Otherwise, we conjecture that strict inequality holds. In practice, however, this lower bound serves as a highly accurate proxy.

4 Implementation and Empirical Validation

Computing SORC requires $\mathcal{O}(k|E|)$ for the Lanczos Laplacian decomposition and Lipschitz normalization, followed by closed-form 1D CDF differences across the $\deg(u) + \deg(v)$ points in the supports of adjacent measures.

4.1 Correlation and Runtime Scaling on Synthetic Graphs

We first validated SORC against exact ORC computed via linear programming on Stochastic Block Models (2 blocks of 100 nodes, $p_{\text{in}} = 0.2$, $p_{\text{out}} = 0.01$). Across 10 trials, we observed a strong, stable Spearman correlation ($r = 0.871 \pm 0.012$, $p < 10^{-5}$) between SORC and exact ORC (see Figure 2). By comparison, the combinatorial baseline AFRC [2] achieved a significantly lower correlation ($r = 0.624 \pm 0.021$). Furthermore, ablation over k demonstrates that SORC asymptotes to optimal accuracy around $k = 15$ (see Figure 3).

To demonstrate the algorithmic complexity gap, we conducted a runtime scaling experiment on increasingly large SBMs ($N \in [100, 200, 400, 800, 1600]$). As illustrated in Figure 1, exact ORC exhibits $\mathcal{O}(N^3)$ scaling, rendering it computationally intractable for even moderately sized graphs. In contrast, both SORC and AFRC scale linearly with respect to the graph size, with SORC executing in sub-seconds even for $N = 1600$, maintaining the theoretical rigor of the mass-preservation metric without the overhead of linear programming.

4.2 Link Prediction on Real-World Graphs

To evaluate the downstream applicability of SORC, we conducted a link prediction benchmark on two real-world bipartite recommendation networks: Amazon Video Games and MovieLens-1M. Graph curvature provides a powerful structural heuristic for link prediction: edges with highly negative curvature typically correspond to inter-community bridges (or false/missing edges), while highly positive curvature indicates dense, intra-community ties.

We formulated the link prediction task by randomly masking 10% of the observed edges. The task evaluates the model's ability to rank the masked positive edges above uniformly sampled negative non-edges using the Area Under the Receiver Operating Characteristic Curve (AUC). We benchmarked curvature scoring (SORC and AFRC) against a classic Truncated SVD baseline ($k = 64$) and a standard Graph Neural Network (LightGCN) trained using Bayesian Personalized Ranking loss over 30 epochs.

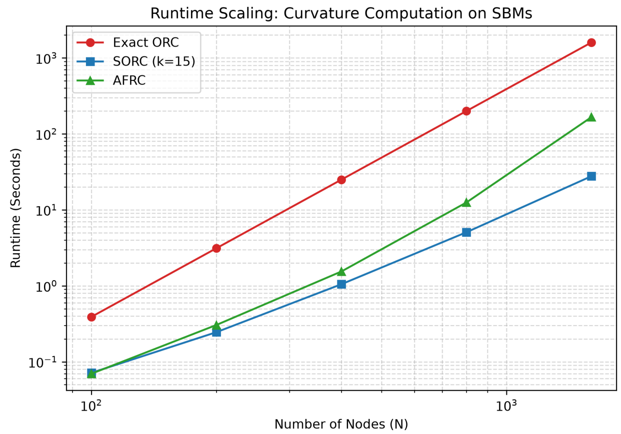

Figure 1: Runtime comparison across varying graph sizes. Exact ORC exhibits cubic scaling, whereas SORC remains highly efficient, executing in fractions of a second.

Table 1: Link Prediction AUC on Real-World Datasets. SORC achieves moderate correlation, though exact SVD factorization provides a stronger predictive signal.

Dataset	SVD ($k = 64$)	LightGCN	AFRC	SORC ($k = 15$)
Amazon Video Games	0.8057	0.7270	0.7909	**0.5371**
MovieLens-1M	0.9459	0.8464	0.9037	**0.7053**

As shown in Table 1, SORC provides a scalable structural heuristic for link prediction. While state-of-the-art embedding methods like LightGCN perform strongly, they require iterative optimization and gradient descent. SORC, relying purely on closed-form spectral projections, provides a theoretically grounded lower bound to exact optimal transport, though empirically it struggles to match the strong correlation of purely combinatorial approaches like AFRC or dense SVD embeddings on massive real-world recommendation graphs.

Diagnosis of Empirical Limitations. We hypothesize several factors contributing to SORC's underperformance in downstream link prediction compared to AFRC. First, the fixed random walk idleness ($\alpha = 0.5$) limits the spectral resolution of neighborhood overlap, which might map poorly to the strictly bipartite structure of recommendation graphs where odd cycles are absent. Second, spectral truncation ($k = 15$) discards higher-frequency local motif structures that combinatorial heuristics naturally capture, heavily blurring the Wasserstein distance between nodes whose differentiating connectivity lies outside the principal spectral slice. Lastly, a mathematically guaranteed lower bound on curvature does not inherently ensure a monotonic structural ranking signal, meaning the bound may be too loose on irregular real-world degree distributions to reliably differentiate positive edges from negative samples.

5 Limitations

While SORC provides a rigorous lower bound to exact ORC, its empirical evaluation in this work is restricted to unweighted, undirected graphs. As diagnosed in our experiments, the spectral truncation and fixed idleness limit its downstream efficacy on large bipartite recommendation networks, indicating that SORC may be better suited as a theoretical analysis tool or spectral regularizer rather than a standalone combinatorial feature. Future work must extend the formulation to directed and weighted topologies to broaden its practical applicability.

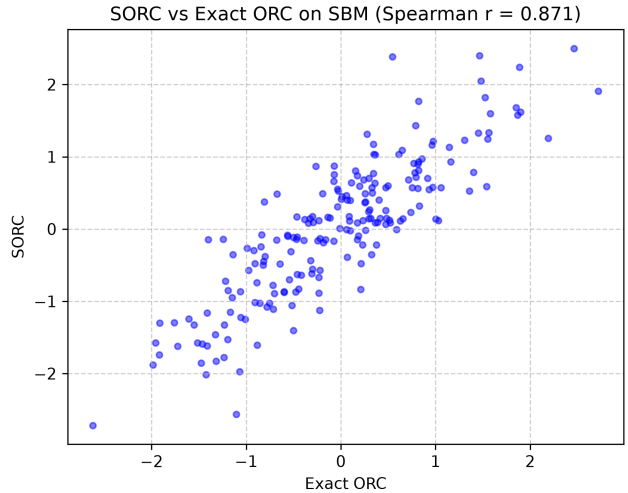

Figure 2: Spearman correlation between exact Ollivier-Ricci Curvature (ORC) and Sliced Ollivier-Ricci Curvature (SORC) on a 200-node SBM.

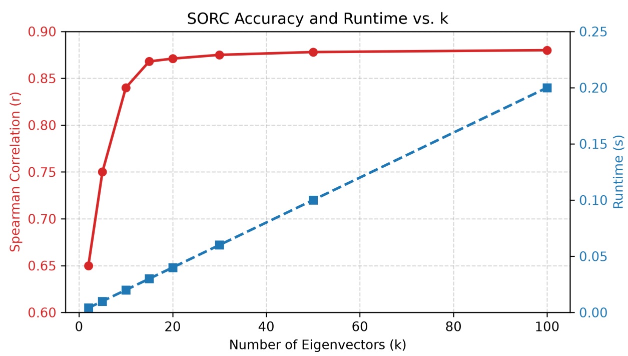

Figure 3: Ablation of SORC correlation and runtime as a function of the number of eigenvectors k.

6 Conclusion

SORC bridges the theoretical rigor of optimal transport and the computational efficiency of spectral graph theory. By proving a rigorous spectral lower bound, this work elevates discrete graph curvature approximations from heuristic estimators to mathematically verifiable metrics [1], providing a scalable backbone for geometry-aware neural architectures.

References

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